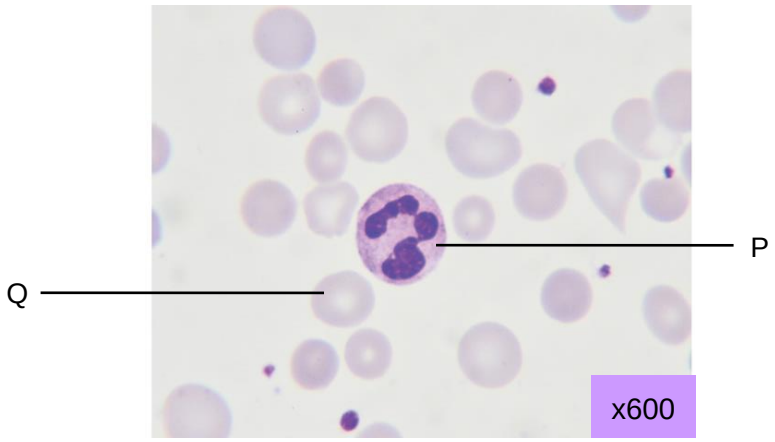
Structured questions

|!|BCELB0090000001|!| ★ **Cross topic**

The photomicrograph below shows some cells of a human blood smear.



(a) (i) The central part of cell Q is lighter in colour. Explain this with reference to its shape. (2 marks)

(ii) What is the significance of this shape for cell Q to carry out its function? (2 marks)

(b) With reference to the photomicrograph, state ***one*** structural difference between cell P and cell Q. What is the significance of such difference to cell Q? (3 marks)

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(c) A woman took a blood test. The report showed that the number of cell Q is less than normal. Name this disease and suggest the possible cause of this disease. (2 marks)

(d) Name the fluid that holds cells P and Q. State ***one*** difference in composition between this fluid and tissue fluid. (2 marks)

##

(a) (i) Cell Q has a biconcave disc shape. (1)
The central region contains less cytoplasm / haemoglobin. (1)
(ii) This provides a large surface area to volume ratio / shortens diffusion distance (1)
to facilitate the diffusion of oxygen into and out of the cell. (1)
(b) Cell P has a nucleus but cell Q does not. (1)
Cell Q can have more space to accommodate more haemoglobin (1)
to increase the oxygen carrying capacity of the cell. (1)
(c) anaemia (1)
Her diet is deficient in iron. (1)
(d) plasma (1)
Plasma contains plasma proteins while tissue fluid does not. (1)

(11 marks)

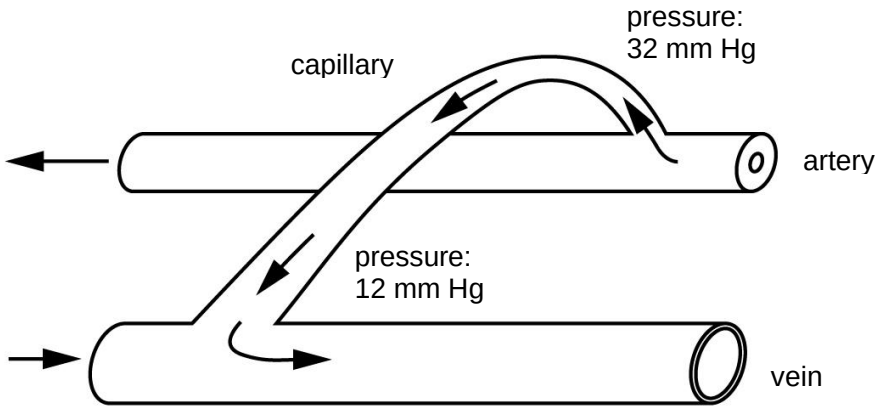
##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	1
	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Understanding basic concepts	2

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|!|BCELB0090000002|!| ★

The simplified diagram below shows the relationship between an artery, vein and a capillary. (The diagram is ***not*** drawn to scale.)



(a) What happens to the pressure after blood flows through the capillary? Explain your answer. (3 marks)

(b) Give ***two*** differences between the artery and veins shown in the diagram. (2 marks)

(c) Capillaries are the site exchange of materials between the blood and body cells. Give ***two*** substances that have a net movement from the blood to the body cells. (2 marks)

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(d) Name the fluid which fills the space between the body cells. How can this fluid return to the heart? (5 marks)

##

(a) The blood pressure decreases. (1)

The blood has overcome great friction / resistance in the capillary. (1)

There is a loss of fluid from the blood during the formation of tissue fluid. (1)

(b) Any ***two*** of the following: (2)

- Arteries carry blood at a higher pressure than veins. (1)
- Arteries have thicker walls than veins. (1)
- Veins have larger lumens than arteries. (1)

(c) oxygen, glucose, amino acids, hormones (any ***two***) (2)

(d) Tissue fluid (1)

Most of the tissue fluid returns to the blood at the venous end of the capillary bed (1)

by osmosis. (1)

The remaining drains into lymph capillaries, which join to form lymph vessels. (1)

Lymph vessels return the lymph to the heart via two large veins. (1)

(12 marks)

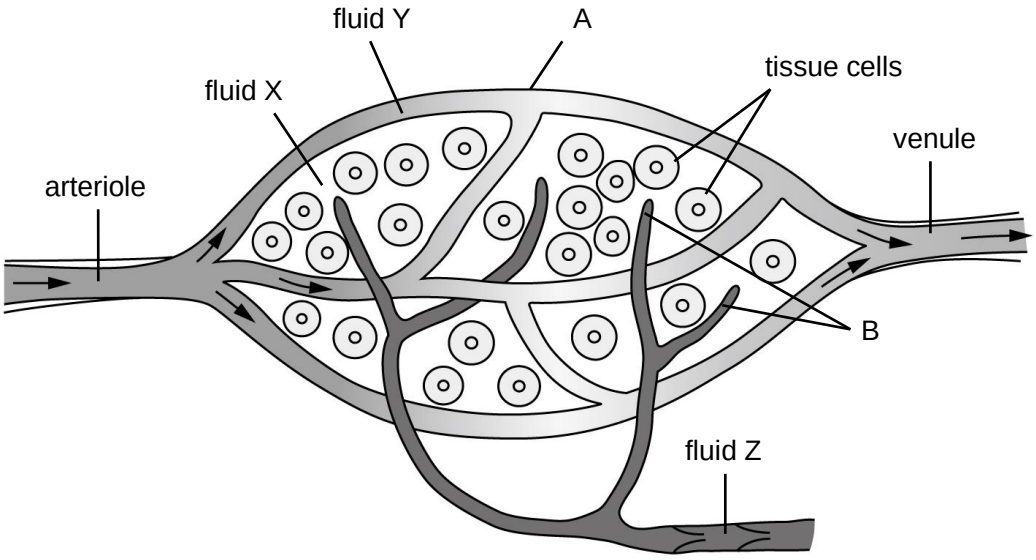
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Analysing data	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	5

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|!|BCELB0090000003|!| ★

The diagram shows part of the circulatory system and its neighbouring cells in the human body. The arrows show the direction of fluid flow.



(a) A is in the form of highly branched network. Explain how this feature is important for A to perform its function efficiently. (4 marks)

(b) (i) Name fluid X and fluid Y. (2 marks)

(ii) Describe the mechanism by which fluid X is formed from fluid Y. (3 marks)

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(c) State ***one*** structural difference between vessel A and vessel B. (1 mark)

(d) Fluid Z will eventually flow back to the heart. Name fluid Z and describe how it continuously flows back to the heart. (3 marks)

##

(a) The large number of branches provide a large surface area (1)
for rapid exchange of materials between the blood and the tissue cells. (1)
The network has a large total cross-sectional area (1)
to reduce the rate of blood flow / to allow more time for the exchange of materials. (1)

(b) (i) fluid X: tissue fluid (1)
fluid Y: blood (1)
(ii) The hydrostatic pressure of fluid Y near the arteriole end is higher than that of fluid X. (1)
Some of the blood / plasma except large proteins / red blood cells / blood platelets (1)
are forced through the thin wall of A to form fluid X. (1)

(c) Vessel A has openings at both ends while B has opening at one end only. (1)

(d) fluid Z: lymph (1)
The contracting of surrounding skeletal muscles helps push fluid Z along the lymph vessels. (1)
OR
During inhalation, the decrease in pressure of the thoracic cavity helps draw fluid Z towards the heart. (1)
The presence of valves along the lymph vessels prevents the backflow of fluid Z. (1)

(13 marks)

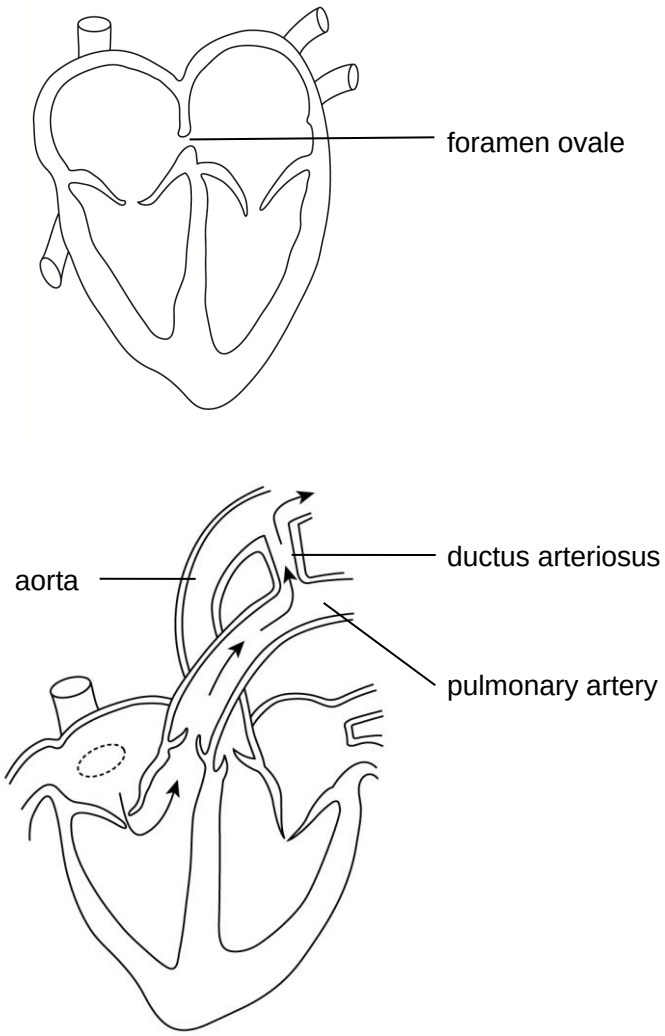
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##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)(i)	Analysing data	2
(b)(ii)	Understanding basic concepts	3
(c)	Analysing data	1
(d)	Understanding basic concepts	3

|!|BCELA0090000004|!| ★★

A foetus receives oxygenated blood directly from its mother, and the circulatory system of a foetus is different from that of a child after birth. In the heart of a foetus, there is a small opening, called the foramen ovale, in the septum. Also, there is a small blood vessel, called ductus arteriosus, connecting the aorta and pulmonary artery.



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(a) State the two heart chambers connected by the foramen ovale. (2 marks)

(b) Explain briefly why foramen ovule and ductus arteriosus are important in foetal blood circulation. (4 marks)

(c) Some babies are born with a heart defect and their foramen ovale remains open. Suggest the effects of this heart defect. (2 marks)

##

(a) right atrium and left atrium (2)

(b) Foramen ovale allows blood to pass from right atrium to left atrium. (1)
Ductus arteriosus allows blood pass from pulmonary artery to aorta. (1)
Since the foetus receives oxygenated blood from its mother, (1)
these two structures allow the oxygenated blood to bypass lungs / to be taken to body tissues. (1)

(c) Deoxygenated blood mixes with oxygenated blood (1)
Less oxygen is delivered to body tissues. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	4
(c)	Applying concepts	2

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|!|BCELA0090000005|!| ★★

Read the passage below and answer the questions that follow.

> Varicose veins are twisted, enlarged veins due to the accumulation of blood. The veins most commonly affected are those in the legs.
>
> Weak or damaged valves can lead to varicose veins. Muscle contraction in the lower legs help blood return to the heart. However, if the valves in the veins are weak or damaged, blood can flow backward and accumulate in the veins, causing varicose veins.
>
> One form of treatment is to wear compression stockings. These stockings are tightest at the ankle and get gradually looser further up the leg.

(a) What is the function of valves in the veins? (1 mark)

(b) Describe the cause of varicose veins. (2 marks)

(c) Why do varicose veins mostly occur in the legs? (2 marks)

(d) People having varicose veins usually feel tired at the legs. Explain this. (2 marks)

(e) Suggest why wearing compression stockings can be a treatment for varicose veins. (2 marks)

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##

(a) To prevent the backflow of blood. (1)

(b) The valves in the veins are weak or damaged. (1)
Blood cannot flow back, leading to accumulation of blood. (1)
The veins become larger as a result.

(c) The legs are far away from the heart (1)
and the veins in the legs must work against gravity. (1)
The blood will more likely to accumulate in the veins if the valves are weak or damaged.

(d) Metabolic wastes like carbon dioxide and lactic acid produced (1)
by the cells in the legs are not removed properly. (1)

(e) Compression stockings create pressure in the lower legs, (1)
which prevents blood from accumulating in the veins and reduces swelling. (1)

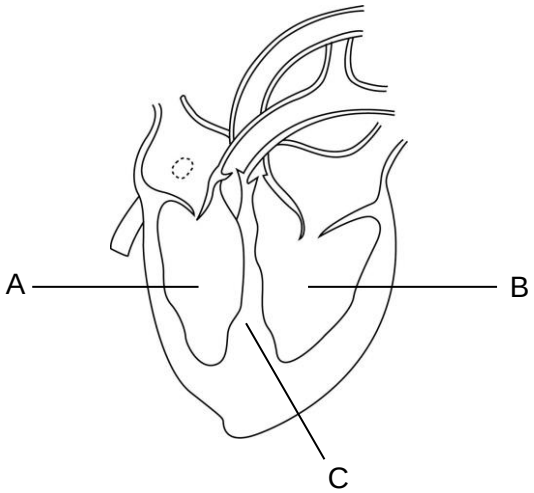
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Applying concepts	2
(e)	Applying concepts	2

|!|BCELB0090000006|!| ★

Flora dissects a rabbit's heart. She found that there are four chambers in the heart. She draws a sketch of the ventral view of the heart as shown below.



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(a) State and explain the difference in thickness of the walls of chambers A and B. (4 marks)

(b) With reference to the diagram above, name ***two*** major blood vessels of the heart that contain deoxygenated blood. (2 marks)

(c) State the heart structures that prevent the backflow of blood. (1 marks)

(d) Some animals do not have structure C. How would this affect these animals? (3 marks)

##

(a) The wall of chamber B is thicker than that of chamber A. (1)
The left ventricle has to produce a greater blood pressure to pump blood (1)
over long distances throughout the body, (1)
It has a thicker wall than the right ventricle which pumps blood only to the lungs. (1)

(b) vena cava (1)
pulmonary artery (1)

(c) valves (1)

(d) Structure C (septum) separates the left and right sides of the heart. (1)
The absence of structure C allows the mixing of oxygenated and deoxygenated blood. (1)
Hence, body cells may obtain less oxygen. (1)

(10 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	3

|!|BCELB0090000007|!| ★

Read the passage below and answer the questions that follow.

Endurance

There is a method to increase the endurance of an athlete. One litre of blood is removed from an athlete and is stored for about six weeks in a freezer. The frozen blood is then thawed and returned (re-infused) to the athlete's bloodstream.

After the removal of blood, the athlete's body tends to produce more red blood cells. After about five weeks, the number of red blood cells in the body will return to normal. Then the thawed blood is returned to the athlete's body. Thus the total number of red blood cells is increased.

(a) State the function of red blood cells. (2 marks)

_____

(b) (i) State the body part which forms red blood cells. (1 mark)

_____

(ii) State the necessary mineral for forming red blood cells. (1 mark)

_____

(c) (i) State the change in oxygen carrying capacity of the athlete after the thawed blood is returned to the body. (1 mark)

_____

_____

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(ii) Describe how this change enables the athlete to get a better performance.

(4 marks)

##

(a) Red blood cells contain haemoglobin (1)
which carries oxygen to all parts of the body. (1)

(b) (i) Bone marrow (1)

(ii) Iron (1)

(c) (i) The oxygen-carrying capacity of the athlete increases after the thawed blood is returned to the body. (1)

(ii) The blood carries more oxygen to the skeletal muscle (1)
for aerobic respiration. (1)
A larger amount of energy is released. (1)
Muscle contraction can occur faster and more powerful, resulting in better performance. (1)

(9 marks)

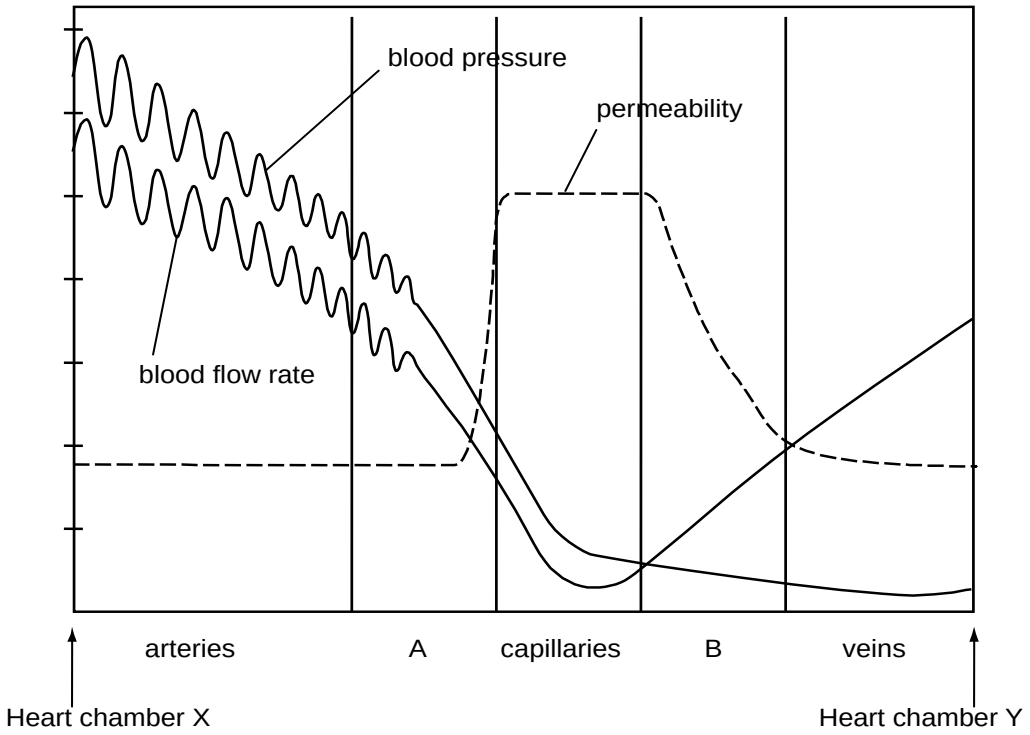
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Applying concepts	4

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|!|BCELB0090000008|!| ★

The graph below shows the changes in the blood pressure, blood flow rate and permeability along different types of blood vessels.



(a) Name heart chambers X and Y. (2 marks)

(b) Name blood vessels A and B. (2 marks)

(c) Describe and explain the changes in blood pressure in the arteries. (3 marks)

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(d) Compare the permeability of the capillaries with those of other types of blood vessels and account for its significance in the exchange of materials. (3 marks)

(e) Describe and explain the blood pressure in veins. (2 marks)

##

(a) X: left ventricle (1)
Y: right atrium (1)
(b) A: arterioles (1)
B: venules (1)
(c) The blood pressure drops gradually from the highest level. (1)
The pressure fluctuates less along the arteries. (1)
This is because the blood is gradually getting further away from the pumping action of the heart. (1)
(d) The permeability of the capillaries is the highest among the blood vessels. (1)
This is because the wall of capillaries is only one-cell thick. (1)
This shortens the distance for the diffusion of different substances (e.g. oxygen, glucose and metabolic wastes) into and out of the capillaries, facilitating the exchange of materials. (1)
(e) The blood pressure drops to the lowest level. (1)
This is because the blood in the venules and veins has overcome great resistance after travelling a long distance. (1)

(12 marks)

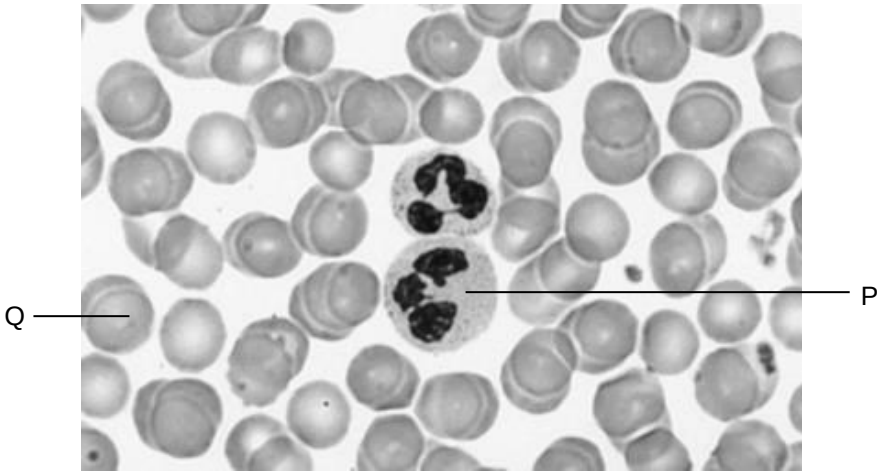
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)	Analysing data	1
	Applying concepts	2
(d)	Analysing data	1
	Applying concepts	2
(e)	Analysing data	1
	Applying concepts	1

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|!|BCELC0090000009|!|

The following photomicrograph shows the appearance of a human blood smear under the light microscope.



(a) Name cells P and Q. (2 marks)

(b) Based on the structures, shapes and functions of the cells, state ***three*** differences between cells P and Q. (3 marks)

(c) (i) Where will cell Q be destroyed when it becomes aged? (1 mark)

(ii) State ***one*** chemical product formed during the breakdown of cell Q. How is this product excreted from the body? (2 marks)

##

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(a) Cell P: White blood cell / phagocyte (1)

Cell Q: Red blood cell (1)

(b)

	P (white blood cell/ phagocyte)	Q (red blood cell)	
Structure	has nuclei, but contains no haemoglobin	has no nucleus, but contains haemoglobin	(1)
Shape	irregular shape	biconcave disc-shaped	(1)
Function	engulfs and digests invading pathogens	contains haemoglobin which carries oxygen from the lungs to all body cells	(1)

(c) (i) Q (red blood cell) will be destroyed in the spleen / the liver. (1)

(ii) Bile pigment is formed during the breakdown of cell Q. (1)

It is excreted from the body in the bile. (1)

(8 marks)

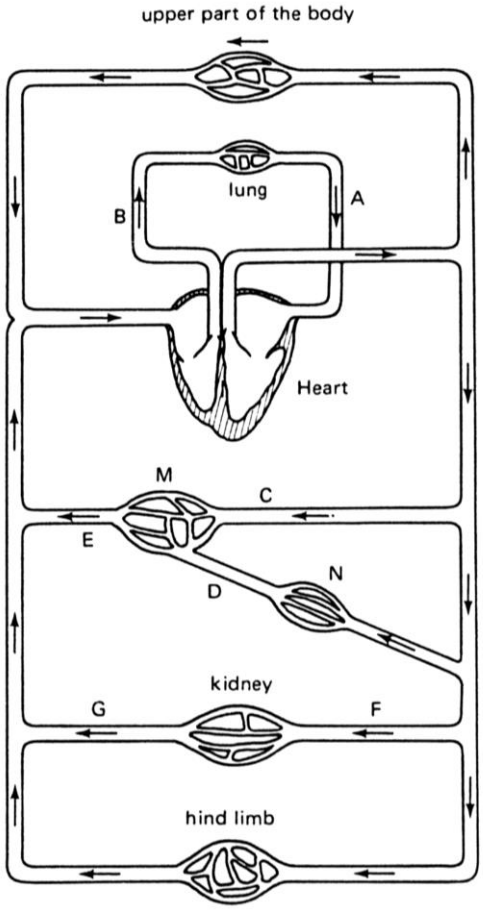
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	3
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

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|!|BCELC0090000010|!| **Cross topic**

Study the plan of the human circulatory system below and answer the questions below.



(a) Using the letters in the diagram, state

(i) the artery which carries deoxygenated blood. (1 mark)

_____

(ii) the organ where digested food substances is absorbed. (1 mark)

_____

(iii) the organ where urea is formed. (1 mark)

_____

(b) Both blood vessels C and D carry blood to organ M. State ***two*** differences between the blood content in these blood vessels several hours after a meal. (2 marks)

_____

_____

_____

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(c) What is the difference between the glucose level of the blood in D and E of a starved man? Explain your answer. (3 marks)

##

(a) (i) B (1)

(ii) N (1)

(iii) M (1)

(b) Any ***two*** of the following: (1 mark each) (2)

C	D
- higher oxygen content	- lower oxygen content
- lower carbon dioxide content	- higher carbon dioxide content
- less glucose content	- more glucose content

(c) The glucose level of the blood in E is higher than that in D. (1)

This is because less glucose is absorbed in the small intestine (N) of a starved man, (1)

and the liver converts the stored glycogen into glucose which is released into blood vessel E. (1)

(8 marks)

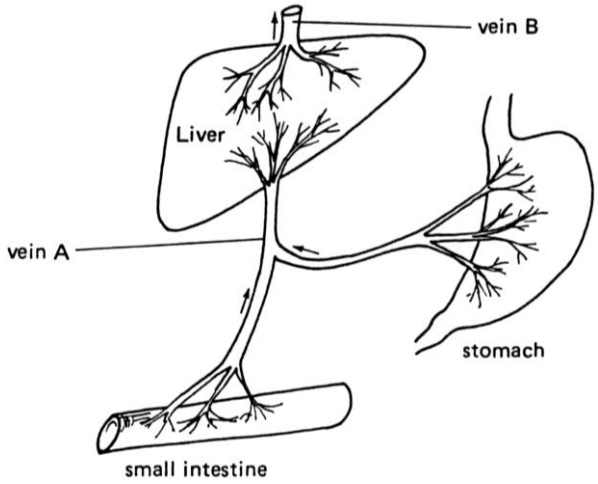
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(a)(iii)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	3

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|!|BCELC0090000011|!| **Cross topic**

The following diagram shows a part of the circulatory system in the human abdominal cavity. The blood vessels shown are veins. The arrows indicate the direction of blood flow.



(a) Name veins A and B. (2 marks)

(b) State ***two*** differences in the compositions between the plasma of veins A and B after a meal. (2 marks)

(c) (i) Name the fluid which is secreted by the liver but not carried in the bloodstream. (1 mark)

(ii) State ***two*** functions of this secretion. (2 marks)

(d) Give the names of ***two*** types of food substances which can be directly absorbed by the stomach. (2 marks)

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##

(a) A: hepatic portal vein (1)

B: hepatic vein (1)

(b) Any ***two*** of the following comparisons or other reasonable answers: (1 mark each) (2)

	Vein A	**Vein B**
Amount of glucose	more	less
Amount of amino acids	more	less
Amount of urea	less	more
Oxygen content	higher	lower
Carbon dioxide content	lower	higher

(c) (i) Bile (1)

(ii) The alkalinity of the secretion neutralizes the acidic chyme. (1)

The bile salts emulsify lipids into smaller oil droplets. (1)

(Or other acceptable answers)

(d) Any ***two*** of the following: (1 mark each) (2)

Water / alcohol / simple sugars / minerals

(Or other acceptable answers)

(9 marks)

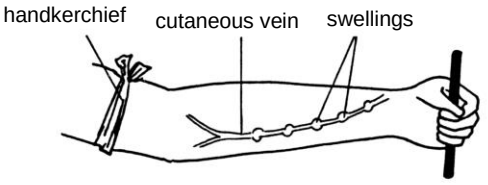
##

	Skills assessed	**Marks**
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

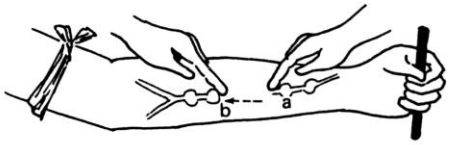
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|!|BCELB0090000012|!| ★

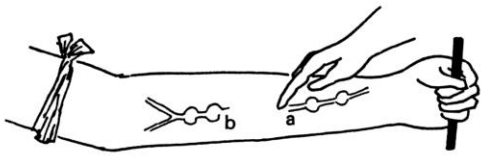
The following diagrams illustrate an experiment performed by the English doctor William Harvey in the 17th century to show the direction of blood flow in veins.



1



2



3

Step 1: A cutaneous vein became conspicuous when the upper arm was tied by a handkerchief.

Step 2: Apply pressure with one finger at site a and squeeze blood along the vein from site a to site b with another finger. The vein between sites a and b became inconspicuous.

Step 3: Remove finger from site b, blood failed to flow back from site b to a.

(a) What do the swellings shown in the above diagrams represent? (1 mark)

_____

(b) Why should the upper arm be tied by a handkerchief before performing the experiment? (1 mark)

_____

_____

(c) Why did the vein between site a and site b become inconspicuous in step 2? (1 mark)

_____

_____

(d) Why did the blood fail to flow back from site b to site a in step 3? (2 marks)

_____

_____

(e) Would the results be different if you remove the finger from site a instead of site b in step 3? (1 mark)

_____

_____

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(f) What can you conclude from this experiment? (1 mark)

##

(a) The swellings represent the positions of the valves in veins. (1)

(b) To slow down the blood from flowing back to the heart, making the cutaneous vein more conspicuous. (1)

(c) The blood was pushed forwards from site a to site b, making the vein between site a and site b inconspicuous. (1)

(d) The valve in site b (1)
prevented the blood from flowing backwards. (1)

(e) Blood would flow from site a to site b. The cutaneous vein between site a and site b thus became conspicuous. (1)

(f) Blood in veins flows in one direction only. (1)

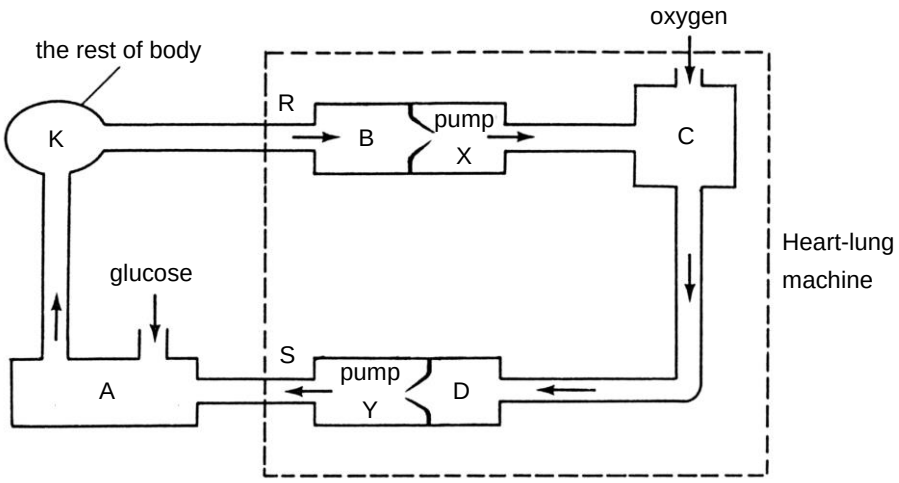
(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
(c)	Applying concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	1
(f)	Applying concepts	1

|!|BCELB0090000013|!| ★ **STSE**

A student constructs a model to represent a 'heart-lung' machine, which is sometimes used when a patient undergoes heart surgery. The model is shown in the following diagram.



(a) Name the structures in the human body that are represented by

(i) pump X and pump Y. (2 marks)

(ii) chambers A and C. (2 marks)

(iii) tubes R and S. (2 marks)

(b) In constructing this model, the student neglects one important function of the 'heart-lung' machine compared with the human transport system. State what the student neglects. (1 mark)

(c) Compare this model with the human transport system again. Can chamber K represent the rest of the body? Explain your answer. (2 marks)

##

(a) (i) Pump X: right ventricle (1)
Pump Y: left ventricle (1)
(ii) Chamber A: small intestine / liver (1)
Chamber C: lung (1)
(iii) Tube R: vena cava (1)
Tube S: aorta (1)

(b) He should set a chamber to remove carbon dioxide and the metabolic wastes from the blood. (1)

(c) No, chamber K cannot represent the rest of the body. (1)
The blood from the digestive system should flow back to the heart first instead of circulating to the rest of the body directly. (1)

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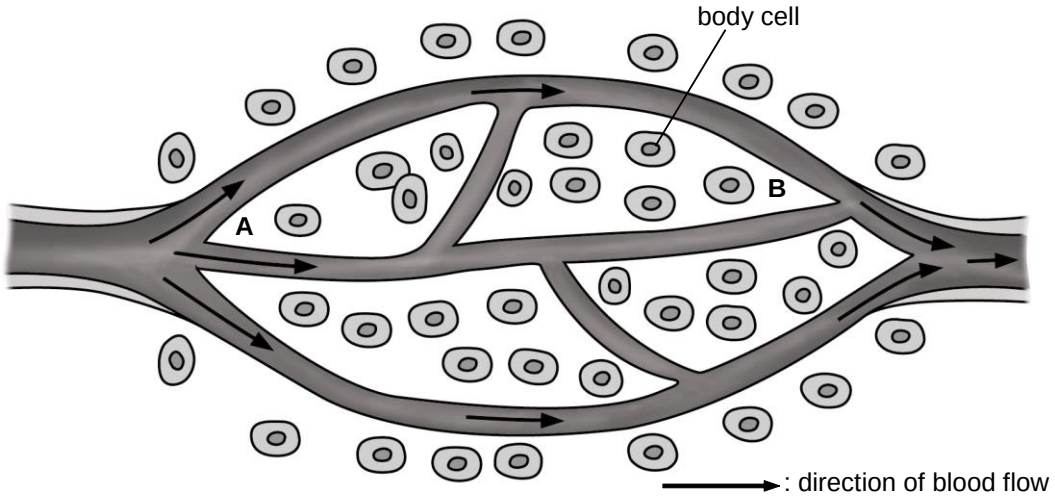
(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Analysing data	2
(a)(iii)	Analysing data	2
(b)	Applying concepts	1
(c)	Applying concepts	2

|!|BCELC0090000014|!|

The following diagram shows some body cells and the capillary bed.



(a) Name ***two*** metabolic wastes that enter the bloodstream from the body cells. (2 marks)

(b) State and explain the difference between the blood at point A and point B in terms of oxygen content. (3 marks)

(c) (i) Name the fluid surrounding the body cells. (1 mark)

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(ii) Explain ***one*** difference in composition between the blood and the fluid named in (c)(i). (2 marks)

##

(a) Carbon dioxide (1)
Urea (1)

(b) The oxygen content in the blood at A is higher than that at B (1)
because oxygen diffuses out of the capillary to the body cells (1)
for respiration to release energy. (1)

(c) (i) Tissue fluid (1)

(ii) The blood contains red blood cells / blood platelets / plasma proteins while the tissue fluid does not. (1)
This is because the red blood cells / blood platelets / plasma proteins are too large to pass through the capillary walls. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2

|!|BCELC0090000015|!|

(a) State ***two*** similarities between veins and lymph vessels. (2 marks)

(b) State ***three*** functions of the lymphatic system. (3 marks)

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(c) Give ***two*** ways by which lymph can be returned to the heart in the correct direction. (2 marks)

(d) Red blood cells are biconcave disc-shaped rather than spherical in shape. Explain how this shape makes it adapt to its function. (2 marks)

##

(a) Valves are present in veins and lymph vessels. (1)
Veins and lymph vessels both carry fluid towards the heart. (1)

(b) Collect excess tissue fluid and returns it back to the blood circulation. (1)
Transport lipids from lacteals to the lymph and then to the blood. (1)
Lymph nodes are packed with white blood cells which destroy pathogens in the lymph. (1)

(c) The valves in the lymph vessels prevent the backflow of lymph. (1)
The flow of lymph is driven by the contraction of muscles surrounding the lymph vessels. (1)

(d) A biconcave disc shape can increase the surface area to volume ratio, (1)
for oxygen to diffuse into and out of the red blood cells more efficiently. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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